

Safety-Constrained Transparent Evaluation of Levee Seepage Stability: An Ordinal Physical Floor for Learned Grades and Evidence-State Control of Inspection Data

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Abstract—Levee seepage assessments increasingly combine physics-based rule chains, data-driven classifiers and unmanned aerial vehicle (UAV) inspection, yet a single safety grade hides whether a learned model has lowered a physically identified risk or whether an inspection candidate was ever confirmed. We formulate transparent evaluation as an auditable decision chain with explicit evidence states, and we test one safety constraint: an ordinal physical floor that lets a random forest (RF) raise, but never lower, the grade of the rule chain. On a controlled benchmark of 875 synthetic scenarios evaluated with grouped five-fold out-of-fold prediction, a threshold-tuned RF raised accuracy from 0.845 to 0.873, but every critical-class record it missed became a direct $C \rightarrow A$ downgrade. No pair among 10,000 threshold pairs eliminated these errors, because the learned features assign the through-structure scenarios one identical vector at every severity. The floor hybrid reached 0.926 accuracy and 0.912 macro-F1, corrected 71 of the rule chain’s 136 errors without introducing any, and removed all critical-class underestimates. Across five additional generator seeds it retained 0.921 ± 0.004 accuracy and 0.998 ± 0.004 critical-class recall. The mechanism is algebraic: the hybrid underestimates only where both components underestimate and inherits the union of their overestimates. In the field data, all nine reviewed infrared piping candidates were false positives, giving an exact one-sided 95% upper bound of 0.28 for candidate precision; unreviewed detections must therefore not enter a grade as confirmed events. These results hold within one synthetic generator; independent field labels are required before any claim of field accuracy.

Index Terms—Levee seepage, piping, transparent evaluation, physics-guided machine learning, random forest, ordinal safety grade, reject option, UAV infrared inspection, evidence provenance.

I. INTRODUCTION

INTERNAL erosion and overtopping dominate the historical failure record of embankment structures: in a review of large embankment dams, piping through the embankment, the foundation or their interface accounted for close to half of the failures and overtopping for a similar share [1], and backward erosion piping remains a governing mechanism for levees on sandy foundations [2]. Operators do not act on these mechanisms directly. They act on a safety grade that compresses soil stratigraphy, hydraulic boundary conditions,

local gradients, slope stability and visible or thermal signs of abnormal discharge into one ordinal class.

Two developments make that compression harder to audit. First, levee digital twins now combine three-dimensional geology, numerical seepage fields and airborne inspection in one scene [3], [4], so a grade can be produced from evidence that is stale, spatially misregistered or merely suspected. Second, data-driven classifiers are increasingly used beside or instead of rule-based indices [5], [6]. A classifier can improve aggregate accuracy while changing the direction of its errors, and in an ordinal safety grade the direction matters: issuing the lowest-risk grade A to a critical case C is not equivalent to issuing B to a safe case.

Existing work addresses parts of this problem. Reliability methods update piping probabilities with field performance observations [7]–[9]; evidence theory combines partial support and ignorance [6], [10]; theory-guided and interpretable machine learning argue that scientific knowledge should constrain learned models in high-stakes settings [11], [12]; and cost-sensitive learning and reject options trade errors against costs or abstention [13]–[15]. What remains unresolved is how a learned grade should be *composed* with a physics-based grade so that the composition cannot produce a false-safe downgrade, what structural property determines whether the composition helps, and how unconfirmed inspection evidence should enter that composition.

This paper asks three questions. (Q1) Under leakage-controlled validation, does a nonlinear learned comparator, here a random forest (RF), improve on a physics-based rule chain, and in which direction do its errors move? (Q2) Can a monotone physical lower bound remove false-safe errors while retaining the learned improvements, and when does that guarantee hold? (Q3) How should algorithmic inspection candidates and incomplete engineering evidence be represented so that they cannot silently change a grade? Our contributions are as follows.

- An evidence-state and admissibility model, together with a versioned operational decision protocol, that separates missing, inapplicable, candidate, reviewed and admissible evidence and records the provenance of every value that reaches a grade (Section III).
- A leakage-controlled benchmark protocol with error-direction metrics, and an analysis showing that the false-safe errors of the learned layer arise from a feature-set

Internal working manuscript, revision of October 2026. Corresponding author: to be completed.

Code, data, per-record predictions and figure sources are in the accompanying de-identified repository (Section VII).

blind spot that no threshold choice can remove (Sections IV and V-B).

- An ordinal physical floor, a proposition that characterizes its error sets, and its evaluation on the original batch, five additional generator seeds, a floor-scope ablation and a review-budget sweep (Sections III-E, V-C and V-D).
- A field audit that treats unmanned aerial vehicle (UAV) infrared detections and a delivered engineering case as evidence-state and provenance tests rather than as accuracy tests (Sections V-F and V-G).

The controlled benchmark and the additional seeds come from one synthetic generator. The results therefore establish the behaviour and the failure conditions of the decision protocol, not field-wide predictive accuracy.

II. RELATED WORK

A. Physical and numerical evidence for seepage safety

Implicit geological modelling interpolates stratigraphic interfaces from boreholes and sections, and the interpolation method and conditioning data are inseparable from the resulting geometry [16], [17]. Transient seepage is commonly described by Richards-type formulations [18], [19] with unsaturated properties such as the van Genuchten relationship [20], and piping models relate local gradients at the exit to erosion progression [2]. Reliability and Bayesian updating propagate parameter variability and incorporate proven-strength observations [5], [7]–[9]. These methods produce physical responses and probabilities, but they do not specify when a response extracted from a particular material zone, scenario and version is admissible input to an operational grade. That admissibility rule is one of the gaps addressed here.

B. Digital twins and provenance

Digital-twin studies describe a loop linking sensing, model updating, simulation and decision support [3], and building information modelling has been combined with mechanism algorithms for the full life cycle of levees [4]. Provenance standards such as PROV-DM model how entities derive from activities and agents [21]. Synchronizing a virtual levee with its physical counterpart does not, however, decide whether a stale, unregistered or merely suspected observation may constrain a safety decision. We therefore use the twin as context and define admissibility separately.

C. Learned safety classifiers and physical guidance

Data-driven seepage evaluation has used surrogate classifiers for reliability analysis [5] and deep learning combined with Dempster–Shafer fusion for dam seepage [6]. Tree ensembles [22]–[24] are attractive comparators because they model interactions among heterogeneous indicators and admit feature-level inspection [25]. Theory-guided data science and the case for interpretable models in high-stakes decisions both argue that scientific knowledge should constrain the learner [11], [12]. Most such approaches constrain the model’s inputs, loss or structure. Few constrain the *direction* of its errors at the decision layer, which is what an ordinal safety grade with asymmetric error costs requires [13].

D. Inspection candidates and selective decisions

UAV photogrammetry and thermography can cover long levee reaches, provided that georegistration, radiometric calibration and flight metadata support the intended inference [26]. Segmentation and detection networks [27], [28] and thermographic deep-learning frameworks [29] can localize seepage-like signatures, but water boundaries, vegetation, shadow and wet ground produce similar thermal contrast. Selective classification and reject options formalize abstention when confidence is low [14], [15]. In levee practice, abstention means sending a record to an engineer, so the review rate and the consequences of the fallback must be measured rather than assumed.

III. TRANSPARENT EVALUATION FRAMEWORK

Fig. 1 summarizes the decision chain. Evidence passes an admissibility gate, the physics-based rule chain and a learned layer each produce a grade, an ordinal physical floor combines them, and an optional review gate routes low-confidence records to an engineer. Every output carries the triggered gates, the evidence states and the version identifiers used to produce it.

A. Evaluation unit, evidence states and admissibility

An evaluation record is indexed by (s, w, t, v) : levee section s , hydraulic scenario w , observation or simulation time t , and evidence-and-rule version v . Each indicator retains its raw value, unit, transformed score, source location, value type (direct measurement, numerical output, representative assignment, controlled synthetic value or manual assignment), applicability, availability, time window, spatial support and transformation rule.

Applicability and availability are separate fields. An indicator is *inapplicable* only after the corresponding facility or failure mode has been confirmed absent; it is *missing* when it should be evaluated but evidence was not obtained. Neither state is encoded as a zero or as a perfect score. Inspection evidence has three further states: *candidate* (algorithmic detection), *reviewed* (manual status recorded) and *admissible* (time, location and status closed for the current unit). Only an admissible confirmed event can impose a deterministic piping or defect constraint. Table I lists the channels and their admissibility conditions.

The evidence table is implemented as a directed provenance graph in the sense of PROV-DM [21]. Nodes represent observations, geological objects, numerical fields, review actions and decision outputs; edges record derivation, spatial registration, temporal precedence and rule triggering. For a unit (s, w, t, v) , indicator x_j is admissible only when its spatial support $\mathcal{S}_j$ intersects the unit support $\mathcal{S}_s$, its validity interval $\mathcal{T}_j$ contains t , and its version v_j is compatible with v :

$$a_j(s, w, t, v) = \mathbf{1}\{\mathcal{S}_j \cap \mathcal{S}_s \neq \emptyset\} \mathbf{1}\{t \in \mathcal{T}_j\} \mathbf{1}\{v_j \preceq v\}. \quad (1)$$

If $a_j = 0$, the value is retained for traceability but cannot become a normal score. A misregistered inspection box or a stale simulation output therefore appears as a provenance failure rather than as a numerical zero.

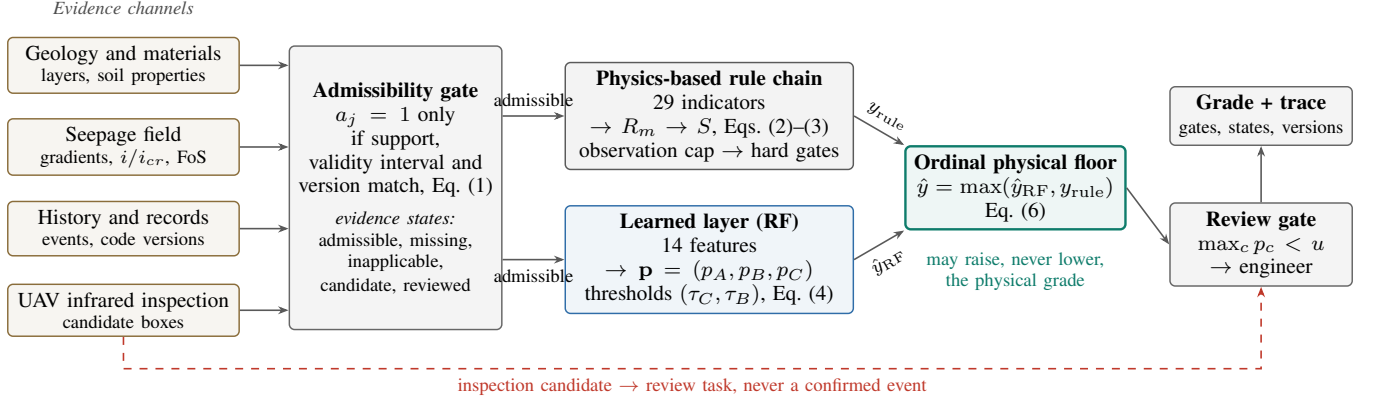


Fig. 1. Transparent evaluation chain (FoS, factor of safety). Evidence enters only through the admissibility gate, which records an explicit state for every indicator. The rule chain and the learned layer each return an ordinal grade (A < B < C in risk). The ordinal physical floor lets the learned layer escalate, but not downgrade, the rule grade. Inspection candidates create review tasks; they constrain a grade only after review closes their time, location and status.

TABLE I
EVIDENCE CHANNELS AND ADMISSIBILITY CONDITIONS.

Channel	Example evidence	Admissibility condition
Geology	control layer, soil properties, contact zone	spatially registered; material identity retained in the numerical domain
Seepage field	local gradient, i/i_{cr} , factor of safety	scenario, boundary condition, extraction region and material known
Inspection	candidate box, confidence, review status	capture time, location and review state stored; candidate $\neq$ confirmed
History	previous piping or seepage event	event interval distinguished from current relevance
Missingness	unavailable or stale input	reported as a state with a missing-item list

B. Physics-based rule chain

The rule chain contains 29 secondary indicators and five failure modes: overtopping, seepage, slope instability, scour and through-structure effects. Let $R_m \in [0, 1]$ be the management-barrier-adjusted response of failure mode m , π_m its normalized weight and α a shortfall coefficient. The composite risk measure and the score are

$$\rho = \alpha \max_m R_m + (1 - \alpha) \sum_m \pi_m R_m, \quad (2)$$

$$S = 100(1 - \rho), \quad (3)$$

with $\alpha = 0.45$. The score grade is A for $S \geq 80$, B for $40 \leq S < 80$ and C for $S < 40$. An observation cap and physical hard gates are then applied in sequence; the gates cover overtopping, the seepage indicators C1, C4 and C5 (foundation, embankment and contact zone), slope factors of safety, scour, structures and settlement. Each stage may keep the grade or move it towards higher risk, never towards lower risk. The dominant failure mode is the largest internal response and is not a calibrated failure probability.

C. Learned comparator

The learned layer uses 14 features from the physical and inspection channels: crest elevation; foundation, embankment

and contact-zone gradients; downstream and upstream slope factors of safety; scour depth; inspection anomaly density and evidence level; the three gradients divided by 0.25; the inverse downstream factor of safety; and the scour depth divided by a 1.0 m reference, which duplicates the raw depth. It never sees target mode, target severity, any delivered grade, the composite score, the risk measure or the dominant mode. The 14 features contain no structure-condition indicator; Section V-B shows the consequence.

Both learners are deterministic NumPy implementations of classification and regression trees (CART) with Gini impurity [22], [24]. Candidate split thresholds are the midpoints between consecutive distinct feature values; when a node holds more than 30 distinct values, the midpoints are taken between 19 quantiles from the 5th to the 95th percentile. The decision tree uses maximum depth 5 and minimum leaf size 5 with all features. The random forest [23] uses 120 bootstrap trees, maximum depth 7, minimum leaf size 3 and four randomly drawn features per split; its class probability is the fraction of trees voting for the class. Cost-weighted variants draw bootstrap samples with probability proportional to a class weight $w_C \in \{1.5, 2, 3, 4, 6\}$ for class C [13]. All hyperparameters were fixed before the additional seeds were generated and were not retuned. The probability vector $\mathbf{p} = (p_A, p_B, p_C)$ is a decision-support score under the controlled target construction, not a calibrated field failure probability.

D. Threshold policy, safety loss and Pareto comparison

For thresholds (τ_C, τ_B) the learned policy is

$$\hat{y}_{RF} = \begin{cases} C, & p_C \geq \tau_C, \\ B, & p_C < \tau_C \text{ and } p_B \geq \tau_B, \\ A, & \text{otherwise.} \end{cases} \quad (4)$$

Because accuracy alone is uninformative when classes are imbalanced and errors are asymmetric [30], policies are compared on accuracy, macro-F1, balanced accuracy, recall for C, the number of C-class underestimates split into C $\rightarrow$ A and C $\rightarrow$ B, the number of A-class overestimates, and the escalation

rate (fraction of records assigned C). To make the asymmetry explicit we use

$$L_{\text{safe}} = 2N(C \rightarrow A) + N(C \rightarrow B), \quad (5)$$

which weights a direct two-level downgrade twice as heavily as a one-level downgrade. We define a policy as non-dominated when no other threshold pair has at least as high macro-F1, balanced accuracy and critical-class recall while having no larger L_{safe} ; escalation rate is retained as an operational resource coordinate rather than hidden inside a single scalar score. The resulting non-dominated policies form the Pareto set [31]. The representative operating point $(\tau_C, \tau_B) = (0.35, 0.20)$ was chosen on a 0.05 grid from 0.20 to 0.90 as the highest macro-F1 point with zero A-class overestimates and C recall of at least 0.85. It is a declared operating point, not an engineering optimum. A 0.01 grid of 10,000 pairs tests whether any threshold alone avoids $C \rightarrow A$ errors.

E. Ordinal physical floor

Encode A, B and C as 0, 1 and 2. Given the learned decision $\hat{y}_{\text{RF}}$ and the final grade y_{rule} of the rule chain, the ordinal physical floor returns

$$\hat{y}_{\text{H}} = \max(\hat{y}_{\text{RF}}, y_{\text{rule}}). \quad (6)$$

We call this policy the *floor hybrid*; tables and figures label it RF + ordinal floor. It retrains neither component and leaves the evidence trace and triggered gates intact. Its error behaviour follows directly from the order structure.

Proposition 1: For a record with reference class y , (i) $\hat{y}_{\text{H}} < y$ if and only if $\hat{y}_{\text{RF}} < y$ and $y_{\text{rule}} < y$; (ii) $\hat{y}_{\text{H}} > y$ if and only if $\hat{y}_{\text{RF}} > y$ or $y_{\text{rule}} > y$; (iii) $\hat{y}_{\text{H}} = 0$ if and only if both components return 0. Consequently, the hybrid's underestimates are the intersection of the components' underestimates, its overestimates are their union, and the hybrid issues a $C \rightarrow A$ decision only where the rule chain does.

Proof: For integers a, b, y : $\max(a, b) < y \Leftrightarrow a < y \wedge b < y$; $\max(a, b) > y \Leftrightarrow a > y \vee b > y$; and $\max(a, b) = 0 \Leftrightarrow a = b = 0$ for $a, b \geq 0$. ■

Proposition 1 states when the floor helps and when it fails. The hybrid gains when the two components err on different records and the learned layer rarely overestimates; it inherits every conservative error of the rule chain; and its protection against $C \rightarrow A$ decisions is exactly as good as the rule chain's. It is not a statement about the accuracy of either component. We also evaluate a *C-only floor* (RF + C-only floor), $\hat{y} = 2$ if $y_{\text{rule}} = 2$ and $\hat{y} = \hat{y}_{\text{RF}}$ otherwise, which removes the rule chain's influence below C. As an operational variant, records with $\max_c p_c < u$ are routed to review. Offline, no reviewer decision exists, so the rule grade is substituted as a proxy; review-rate and fallback metrics are reported separately from the closed-set metrics.

F. Operational decision protocol

The method is executed as a versioned decision protocol. For each evaluation unit, the implementation (i) freezes the key (s, w, t, v) and records the source, spatial support and validity

interval of every input; (ii) assigns an evidence state and excludes values that fail the admissibility test; (iii) evaluates the rule chain and learned layer independently while preserving their grades, probabilities and triggered gates; (iv) applies the ordinal floor and computes safety-loss and escalation diagnostics; and (v) creates review tasks for low-confidence or candidate inspection states before publishing the grade. The output must retain the grade, triggered gates, evidence state, provenance record, model version and review status. The Pareto sweep therefore selects quality–safety trade-offs, while the review budget and admissibility rules determine deployability.

IV. DATA AND VALIDATION PROTOCOL

A. Controlled benchmark

The benchmark was produced by a frozen generator and contains 875 records: seven target modes (safe reference, overtopping, seepage, slope, scour, through-structure and mixed) $\times$ five severities (safe, slight, moderate, severe, critical) $\times$ 25 records. The reference class follows the frozen severity mapping: safe and slight $\rightarrow$ A (350 records), moderate $\rightarrow$ B (175), severe and critical $\rightarrow$ C (350). It is a closed-set reference for testing decision behaviour under the generator, not a field safety label. The rule chain's final grades were reconstructed from the delivered final-grade error trace rather than from the raw grade column of the input file, and a replay of the generator reproduced all five delivered benchmark files byte for byte.

B. Splits and validation hierarchy

The 875 records contain 722 distinct 14-feature vectors. Identical vectors are assigned to the same fold, and all learned results are grouped five-fold out-of-fold (OOF) predictions, which prevents duplicated inputs from leaking across folds [32], [33]. Forest seeds are derived from the fold index and held fixed across methods. A five-outer by three-inner grouped nested validation re-selects (τ_C, τ_B) inside each outer fold (with 80 trees). Validation is organized in three levels: the original 875 records test behaviour against the frozen reference; five additional 875-record batches (generator seeds 20260930–20260934, same construction, features, splits, hyperparameters and policies) test sensitivity to the generator realization; and the inspection and engineering packages test whether evidence states and provenance survive the interface. None of the latter is used to claim external predictive accuracy.

C. Statistical analysis

Paired comparisons with the rule chain on the same records use the exact two-sided McNemar test on per-record correctness [34], [35]. Within-batch uncertainty of the hybrid-minus-rule differences is summarized by a paired percentile bootstrap that resamples the 722 feature-vector groups (2000 resamples) [36]; it conditions on the fitted OOF predictions and does not include retraining variability. Results across the five additional seeds are reported as mean $\pm$ sample standard deviation (SD) without inferential claims, because

five realizations of one generator do not support population inference. For the 0/9 confirmed field candidates we report the exact Clopper–Pearson bound [37].

D. Field inspection package

The inspection package contains 114 field thermal infrared images with GPS and capture-time metadata, nine algorithmic piping candidates from those images that entered manual review, and 13 images of artificial piping with 14 detector boxes. The remaining field images have no image-level truth labels, and the artificial set has no record of missed detections, so neither supports a detection accuracy, recall or F1 estimate.

E. Engineering-case trace

The delivered K45+500 case, a representative section of a reach with a recorded 2017 piping event, is used only to test provenance and interface fields. Its values are delivery excerpts rather than an independent physical re-run, and its input includes the historical piping information, so it is not a blind test.

F. Reproducibility

All analyses use fixed seeds and write their feature manifests, fold assignments, threshold grids, per-record predictions and summary metrics. The supplementary statistics and all data figures in this paper are regenerated by two scripts that read only repository files and reuse the unchanged model code (Section VII).

V. RESULTS

A. The rule chain errs conservatively, and its gates do the work

The rule chain classified 739 of 875 records correctly (accuracy 0.845, macro-F1 0.827, balanced accuracy 0.847; Table II). All 136 errors were one-level errors (Fig. 2a): 54 A→B and 25 B→A decisions and 57 C→B underestimates, with no C→A or A→C decision. The score grade alone was correct for only 450 records; the observation cap changed no grade, and the hard gates changed 322 grades. An ablation of the inspection channel confirmed its inertness in this benchmark: replacing the inspection indicator C3 with its no-anomaly anchor changed 150 scores by at most 1.39 points, but no grade. The rule chain’s accuracy therefore rests on gates that are aligned with the generator’s severity construction, which makes it a strong comparator and is a property of the benchmark rather than evidence of field skill.

B. Learned policies improve aggregate metrics but create false-safe downgrades

The shallow tree underperformed the rule chain on every aggregate metric (Table II). The majority-vote forest raised C recall from 0.837 to 0.854 but lowered accuracy and macro-F1, and all 51 of its C-class misses were direct C→A decisions, doubling L_{safe} from 57 to 102. The representative threshold policy reached 0.873 accuracy and 0.858 macro-F1.

Its accuracy gain over the rule chain was not distinguishable within the batch (125 records correct only under the RF policy against 100 correct only under the rule chain; exact McNemar $p = 0.11$), and all 50 of its C-class misses were again C→A ($L_{\text{safe}} = 100$).

Threshold tuning cannot remove these errors. Of the 10,000 pairs on the 0.01 grid, none produced fewer than 50 C→A decisions; the best pair by macro-F1, (0.19, 0.06), reached 0.890 macro-F1 with the same 50 C→A errors (Fig. 3). Cost weighting removed them only at $w_C = 6$, where accuracy fell to 0.707 and 179 of 350 A records were escalated; weights 1.5 to 4 left 50 or 51 C→A errors. The nested validation reproduced the pattern (accuracy 0.881, macro-F1 0.870, 50 C→A errors in the pooled outer predictions).

The cause is a blind spot in the learned features rather than a poor threshold. All 50 C→A errors occur in the through-structure target mode (Fig. 2d), whose severity is expressed through structure indicators that the rule chain gates on but the 14 learned features do not contain. In that mode all 125 records, from safe to critical, share one identical feature vector, the same vector as the safe reference scenarios. No function of those features can separate them. The learned layer is nevertheless complementary to the rule chain elsewhere: it corrected the 57 C→B errors that the rule chain made in the safe-reference, overtopping and slope modes and the 54 A→B errors in the seepage, mixed and scour modes.

C. The ordinal floor removes underestimates without adding errors

The floor hybrid reached 0.926 accuracy, 0.912 macro-F1, 0.928 balanced accuracy and C recall 1.000, with no C-class underestimate and $L_{\text{safe}} = 0$ (Table II, Fig. 2c). Relative to the rule chain it corrected 71 of 136 errors and introduced none (exact McNemar $p < 10^{-20}$). The paired group bootstrap gave differences of +8.1 percentage points in accuracy (95% interval 5.4 to 11.5), +8.5 points in macro-F1 (6.0 to 11.3) and −57 in L_{safe} (−72 to −42). Relative to the RF threshold policy, the floor changed 154 decisions: it corrected 100 RF errors (all 50 C→A and 50 of 61 B→A) and introduced 54 A→B escalations, which are exactly the rule chain’s conservative errors.

Proposition 1 accounts for each of these counts. The rule chain and the RF policy never missed the same C record, so the intersection of their C underestimates, and hence the hybrid’s, was empty. The RF policy made no overestimate, so the union of overestimates equals the rule chain’s 54. The 11 remaining B→A errors are safe-reference records that both components placed in A. The hybrid’s escalation rate of 0.400 equals the C prevalence of the benchmark.

The scope of the floor matters. The C-only floor reached a higher accuracy (0.930) and avoided the 54 A→B escalations, but its balanced accuracy fell to 0.884 because it discarded the rule chain’s B-level corrections (61 B→A errors remained). More importantly, it does not inherit the C→A guarantee of Proposition 1(iii): a C record that the rule chain places in B and the learned layer places in A is issued A. This occurred in one of the five additional seeds (Section V-D).

TABLE II

GROUPED FIVE-FOLD OOF COMPARISON ON THE 875 CONTROLLED RECORDS. $C \rightarrow A$ AND $C \rightarrow B$ COUNT C-CLASS UNDERESTIMATES (350 C RECORDS); $A \uparrow$ COUNTS A-CLASS OVERESTIMATES (350 A RECORDS); ESC. IS THE FRACTION OF RECORDS ASSIGNED C (REFERENCE PREVALENCE 0.400). THE PROPOSED ORDINAL FLOOR IS SHADED.

	Method	Accuracy	Macro-F1	Bal. acc.	Recall C	$C \rightarrow A$	$C \rightarrow B$	L_{safe}	$A \uparrow$	Esc.
Physics	Rule chain	.845	.827	.847	.837	0	57	57	54	.335
	Decision tree	.707	.646	.635	.631	129	0	258	0	.253
Learned only	RF, majority vote	.829	.784	.763	.854	51	0	102	0	.342
	RF, threshold (0.35, 0.20)	.873	.858	.836	.857	50	0	100	0	.343
	RF, best of 0.01 grid (0.19, 0.06)	.887	.890	.882	.857	50	0	100	24	.361
	RF, C weight 6	.707	.701	.683	1.000	0	0	0	179	.683
	RF + C-only floor	.930	.903	.884	1.000	0	0	0	0	.400
Floor policies	RF + ordinal floor (Eq. 6)	.926	.912	.928	1.000	0	0	0	54	.400
	RF + ordinal floor + review, $u = 0.60$	.911	.896	.914	.966	0	12	12	54	.386

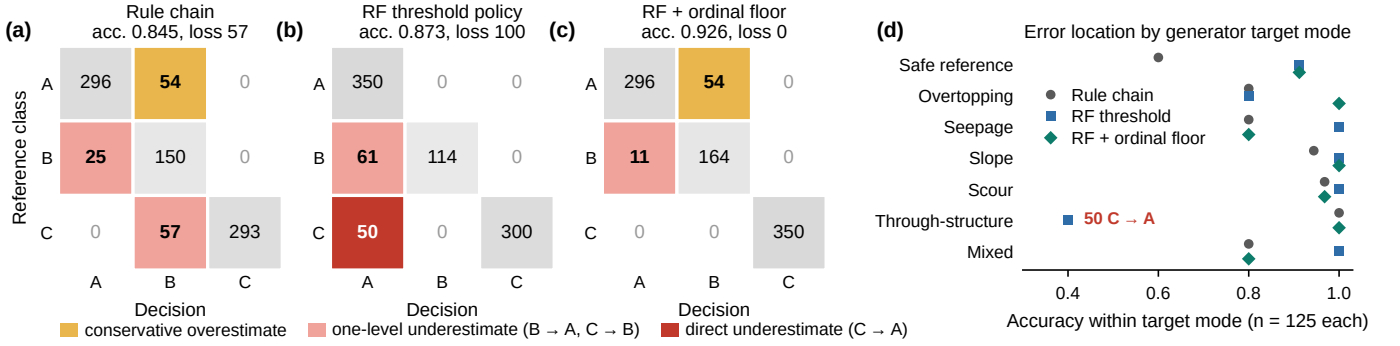


Fig. 2. Error structure on the 875 controlled records. (a)–(c) Confusion matrices for the rule chain, the RF threshold policy $(\tau_C, \tau_B) = (0.35, 0.20)$ and the floor hybrid (RF + ordinal floor); cell colour encodes error direction. The RF policy removes the rule chain’s $C \rightarrow B$ errors but converts its C-class misses into 50 direct $C \rightarrow A$ decisions. The floor hybrid keeps the rule chain’s 54 conservative $A \rightarrow B$ errors and none of the underestimates of either component. (d) Accuracy within each generator target mode ($n = 125$). All RF $C \rightarrow A$ errors occur in the through-structure mode, where every record shares one 14-feature vector.

TABLE III

FIVE ADDITIONAL GENERATOR SEEDS (MEAN $\pm$ SAMPLE SD OVER SEEDS 20260930–20260934; 875 RECORDS EACH). REVIEW RATE IS THE FRACTION OF RECORDS WITH $\max_C p_C < u$, ROUTED TO THE RULE GRADE AS AN OFFLINE REVIEWER PROXY.

Method	Accuracy	Macro-F1	Bal. acc.	Recall C	$C \rightarrow A$	L_{safe}	$A \uparrow$	Review (%)
Rule chain	.847 $\pm$.002	.830 $\pm$.002	.849 $\pm$.001	.842 $\pm$.002	0	55.4 $\pm$ 0.5	53.2 $\pm$ 1.1	–
RF, majority vote	.827 $\pm$.008	.789 $\pm$.013	.767 $\pm$.012	.837 $\pm$.013	57.0 $\pm$ 4.5	114.0 $\pm$ 9.1	0	–
RF, threshold (0.35, 0.20)	.875 $\pm$.010	.862 $\pm$.015	.841 $\pm$.016	.855 $\pm$.004	50.6 $\pm$ 1.3	101.2 $\pm$ 2.7	0	–
RF + C-only floor	.933 $\pm$.010	.907 $\pm$.015	.888 $\pm$.016	.998 $\pm$.004	0.6 $\pm$ 1.3	1.2 $\pm$ 2.7	0	–
RF + ordinal floor	.921 $\pm$.004	.906 $\pm$.005	.920 $\pm$.006	.998 $\pm$.004	0	0.6 $\pm$ 1.3	53.2 $\pm$ 1.1	–
RF + ordinal floor + review, $u = 0.50$	.918 $\pm$.006	.902 $\pm$.007	.917 $\pm$.007	.990 $\pm$.008	0	3.4 $\pm$ 2.8	53.2 $\pm$ 1.1	1.5 $\pm$ 0.7
RF + ordinal floor + review, $u = 0.60$	.908 $\pm$.006	.892 $\pm$.007	.908 $\pm$.008	.969 $\pm$.008	0	10.8 $\pm$ 2.8	53.2 $\pm$ 1.1	6.6 $\pm$ 0.9

Removing the two inspection features from the learned layer changed 23 RF decisions and lowered the hybrid’s accuracy from 0.926 to 0.917 with one $C \rightarrow B$ error. Within the generator the synthetic inspection features thus carried a small amount of signal for the learned layer, whereas they carried none for the rule chain’s grades.

D. Robustness across generator seeds

The pattern held in each of the five additional batches (Table III, Fig. 4a,b). The floor hybrid introduced no new error relative to the rule chain in any seed, corrected 61 to 68 rule errors per seed (exact McNemar $p < 10^{-18}$ in every seed), and improved accuracy, macro-F1 and balanced accuracy over the paired rule chain by 7.4 ± 0.3 , 7.7 ± 0.4 and 7.1 ± 0.5

percentage points. The RF threshold policy’s accuracy gain over the rule chain reached $p < 0.05$ in only one seed (range $p = 0.017$ to 0.34). In seed 20260932 both components missed the same three C records; the floor hybrid issued B for them (three $C \rightarrow B$), whereas the C-only floor issued A (three $C \rightarrow A$), as Proposition 1 predicts. The majority-vote forest’s C recall (0.837 ± 0.013) fell below the rule chain’s (0.842 ± 0.002), so its gain on the original batch did not replicate. The small between-seed spread shows that the decision-layer behaviour is reproducible within the generator family; all batches share the feature construction, severity mapping and rule logic, so the spread says nothing about transfer to field data.

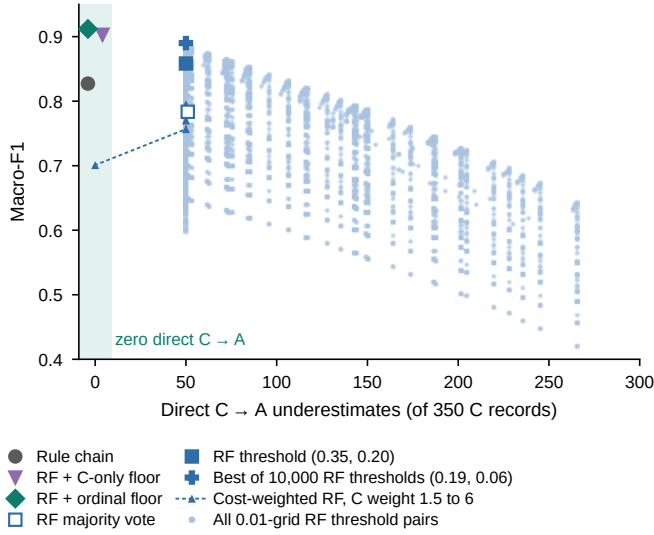


Fig. 3. Quality against direct C→A errors on the 875 controlled records. Light points are all 10,000 RF threshold pairs on the 0.01 grid; no pure RF threshold reaches fewer than 50 C→A errors. The dashed line traces cost-weighted forests ($w_C = 1.5$ to 6). Only the rule chain, the C-weight-6 forest and the two floor policies reach zero, and RF + ordinal floor has the highest macro-F1 among them. Points at zero are offset horizontally by up to four units so that none is hidden.

E. Review budget

Routing low-confidence records to review had a cost that grew with the review rate (Fig. 4c). At $u = 0.50$, $1.5 \pm 0.7\%$ of records were routed, C recall was 0.990 ± 0.008 and L_{safe} was 3.4 ± 2.8 (range 0 to 6); this point met the exploratory budget of at most 5% review and $L_{\text{safe}} \leq 6$ in every seed. At $u = 0.60$ the review rate rose to $6.6 \pm 0.9\%$ and L_{safe} to 10.8 ± 2.8 ; at $u = 0.70$, to $16.0 \pm 1.4\%$ and 37.4 ± 8.6 . On the original batch, $u = 0.50$ routed 7 records and changed no metric, and $u = 0.60$ routed 62 records (7.1%) and produced 12 C→B errors (Table II). These costs measure the proxy, not review: the rule grade substitutes for an engineer, so the sweep quantifies the upward corrections that a reviewer would have to recover. A reviewer that returns the correct reference grade on routed records and is constrained never to grade below the rule chain would remove, rather than create, the proxy errors; the reported sweep therefore measures the cost of substituting the rule grade for an unavailable reviewer. The thresholds were selected post hoc within the same generator family and must be fixed before an external evaluation.

F. Inspection candidates as evaluation states

All nine field candidates that entered manual review were rejected as piping (Fig. 5). Their detector confidences ranged from 0.466 to 0.777. The reviewer attributed five to the thermal contrast at water–land boundaries, two to vegetation and shadow, one to the thermal inertia and shadow of a building edge, and one to wet ground or lens-edge vignetting. With 0 confirmations in 9 reviews, the exact one-sided 95% upper bound on candidate precision is 0.28: the data exclude a high-precision detector, but they do not show that the detector never finds piping. The 14 boxes on the 13 artificial-piping images

TABLE IV
K45+500 DELIVERY CASE: VALUES AND PROVENANCE STATUS.

Item	Delivered value	Provenance status
Upstream water level	35.90 m (crest 36.30 m)	representative scenario, not a measured 2017 level
Controlling layer	silty clay, $i = 0.865$, $i_{cr} = 0.827$, ratio 1.045	numerical output; extraction location and material retained; C1 gate fired
Downstream slope	FoS 1.155 (Bishop simplified)	below the 1.35 allowable value; D1 gate fired
Scour depth	0.40 m	D3 gate fired
Score and grade	24.35; C, seepage-dominant	delivery excerpt; no independent re-run
Completeness	15 of 29 indicators missing	reported as a missing-item list
Consistency	4 table–result conflicts	unresolved (A8, A9, B2, D5)
Code version	module differs from benchmark	adds “confirmed piping → worst C3”; history enters the input

had higher confidences (0.670 to 0.930), but without missed-detection records they establish no recall or field transfer.

The implication for the inspection indicator is direct. A candidate may create a review task, but it cannot enter the grade as a confirmed piping event until time, location and manual status are closed, and an image that was not inspected remains unobserved rather than “no anomaly”. The two highest-confidence field candidates (0.777 and 0.767) were a water edge and vegetation, so a confidence threshold alone would not have separated them from genuine seepage.

G. Engineering-case provenance trace

The K45+500 delivery returned a C grade with a seepage-dominant mode and a score of 24.35 (Table IV). The provenance fields exposed five issues that prevent its use as validation: the water level is a representative high-water scenario rather than a measured 2017 level; 15 of the 29 indicators were reported as missing; four indicators conflict between the delivered indicator table and the result file; the evaluation module used for the case differs from the benchmark version and adds a rule that sets the inspection indicator to its worst value when piping is confirmed; and the confirmed historical piping event is part of the input. The trace therefore demonstrates that the interface carries the information needed to detect these problems; it does not test predictive skill.

VI. DISCUSSION

The safety gain in this paper comes from the composition, not from the learner. A forest trained on the same physical channels improved aggregate metrics only modestly and, on every batch, converted its critical-class misses into direct downgrades; the ordinal floor removed those downgrades and kept most of the learned corrections. Proposition 1 explains why this worked here and why it need not work elsewhere: the floor is valuable exactly when the two components err on different records and the learned layer rarely overestimates. In the benchmark, 125 of the rule chain’s 136 errors fell in the safe-reference, overtopping, seepage and mixed modes, 100

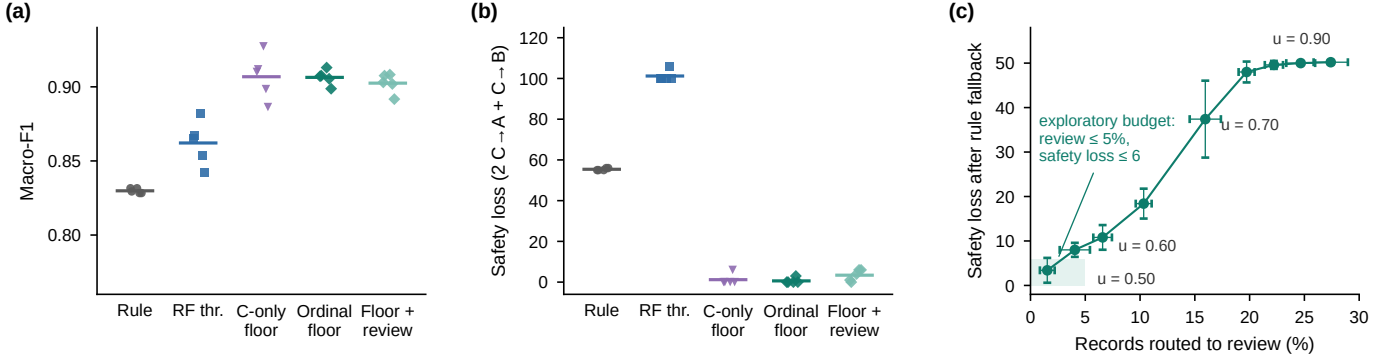


Fig. 4. Five additional generator seeds. (a) Macro-F1 and (b) safety loss L_{safe} per seed (points) and their mean (bar); C-only floor, ordinal floor and floor + review denote RF + C-only floor, RF + ordinal floor and RF + ordinal floor + review ($u = 0.50$). (c) Review-budget sweep for RF + ordinal floor, $u \in \{0.50, 0.55, \dots, 0.90\}$ (mean $\pm$ SD over seeds). Because the offline fallback replaces a routed decision with the rule grade, every routed record can only lose an upward correction, so L_{safe} rises with the review rate. The shaded box marks the exploratory budget of at most 5% review and $L_{\text{safe}} \leq 6$.

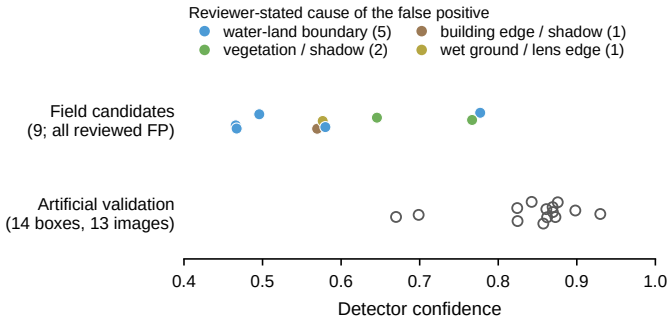


Fig. 5. Detector confidence of the nine reviewed field candidates, coloured by the reviewer-stated cause of the false positive, and of the 14 boxes on artificial-piping images (no review status or missed-detection record). Vertical position within a row is jitter for visibility.

of the forest’s 111 errors fell in the through-structure and overtopping modes, and the two never missed the same C record.

This positions the floor between two existing strategies. Theory-guided learning places physical knowledge in the learner’s inputs, loss or architecture [11], and interpretable models make the learner itself auditable [12]. The floor instead places physical knowledge in the *decision layer* as an ordinal constraint, which leaves the learner unrestricted but bounds the direction of its influence. It is simpler than Bayesian updating of failure probabilities [7], [8] and does not replace it; the two answer different questions. Unlike a reject option [14], [15], the floor never abstains; in our sweep, abstention with a rule-based fallback only removed upward corrections, so the value of review lies in information that the offline proxy cannot represent.

Two rival explanations bound the size of the gain. First, the rule chain’s hard gates are aligned with the generator’s severity construction, so the reference labels partly encode the rule logic; part of the hybrid’s advantage over a pure learner is therefore built into the benchmark. Second, the forest’s $C \rightarrow A$ errors stem from a feature set that omits structure indicators. Adding those indicators would likely shrink the learner’s blind spot and narrow the hybrid’s margin. Neither explanation removes the protective role of the floor, which

does not depend on knowing where the learner’s blind spots are. That property matters for field deployment, where blind spots are unknown in advance.

The inspection results lead to the same design principle at the evidence layer. A detector can be optimized for candidate generation, but the evaluation should consume reviewed event objects with a location, a time window, a confidence and a status. Retaining rejected candidates is useful for retraining and flight planning [29]; promoting them to confirmed piping would have raised the grade of a reach on the basis of water edges and vegetation. The provenance graph keeps that distinction in the data model rather than in the visualization layer.

The claims are bounded in four ways. (i) All quantitative comparisons use one synthetic generator; the five additional seeds test realization variability, not transfer, and the reference labels share the rule logic. (ii) The $C \rightarrow A$ guarantee is conditional on the rule chain: in the field, a rule chain that misses a critical case entirely would pass that error through the floor, and the hybrid inherits every rule overestimate, which in the benchmark meant 54 unnecessary escalations. (iii) The review fallback is a proxy, so the reported review costs are not measurements of engineering review. (iv) The field evidence consists of nine reviewed candidates and one non-blind engineering trace; it supports the evidence-state design but no detection or prediction metric.

Each boundary points to a discriminating test. Independent field labels, timed and registered against inspection and water-level records, would test whether the components remain complementary outside the generator, which Proposition 1 identifies as the condition that matters. Retraining the learner with structure indicators would separate the feature blind spot from the protective effect of the floor. Calibrating the forest’s probabilities and recording actual reviewer decisions would turn the review gate from a proxy into a measured operating point.

VII. CONCLUSION

An ordinal physical floor that lets a learned layer raise but never lower a physics-based grade improved a levee seepage

rule chain without adding a single error relative to that chain in six controlled batches, and removed the direct critical-to-safe downgrades that no threshold or moderate cost weighting of a random forest could avoid. The gain follows from a simple property: the composition underestimates only where both components underestimate, so it helps when their errors are complementary and protects against direct downgrades only to the extent that the rule chain does. Treating inspection detections as candidates until reviewed, and engineering deliveries as provenance traces, kept unconfirmed and inconsistent evidence out of the grade. Field labels with independent timing and registration are the next requirement for testing whether the complementarity, and hence the gain, survives outside the synthetic generator.

DATA AND CODE AVAILABILITY

The de-identified repository contains the controlled benchmark and its generator (data/875_benchmark), the multi-seed batches and analysis scripts (experiments/analysis), the inspection audit tables (inspection/audit) and this manuscript. The statistics and figures specific to this revision, including per-record predictions for the 875 records, are produced by experiments/analysis/paper_revision/paper_statistics.py and make_paper_figures.py, which reuse the unchanged model code. Raw field images were released with metadata removed where possible; locations are not reported in this paper.

ACKNOWLEDGMENT

The authors acknowledge the project team for providing the controlled evaluation package, engineering-case materials and UAV inspection records. Author list and funding information will be completed after internal review.

REFERENCES

- [1] M. Foster, R. Fell, and M. Spannagle, "The statistics of embankment dam failures and accidents," *Canadian Geotechnical Journal*, vol. 37, no. 5, pp. 1000–1024, 2000.
- [2] V. M. van Beek, H. M. van Essen, K. Vandenboer, and A. Bezuijen, "Developments in modelling of backward erosion piping," *Géotechnique*, vol. 65, no. 9, pp. 740–754, 2015.
- [3] A. Fuller, Z. Fan, C. Day, and C. Barlow, "Digital twin: Enabling technologies, challenges and open research," *IEEE Access*, vol. 8, pp. 108 952–108 971, 2020.
- [4] Y. Hou, J. Yu, W. Liu *et al.*, "Research on the full life cycle safety of levee based on BIM+mechanism algorithm," *Archives of Civil Engineering*, vol. 72, no. 2, pp. 247–263, 2026.
- [5] B. Xu, Z. Liu, R. Pang, and Y. Zhou, "Data-driven collaborative safety evaluation for seepage reliability of embankments considering spatial variability," *Journal of Hydrology*, vol. 660, p. 133490, 2025.
- [6] X. Chen, Y. Xu, H. Guo *et al.*, "Comprehensive evaluation of dam seepage safety combining deep learning with Dempster-Shafer evidence theory," *Measurement*, vol. 226, p. 114172, 2024.
- [7] T. Schweckendiek, A. C. W. M. Vrouwenvelder, and E. O. F. Calle, "Updating piping reliability with field performance observations," *Structural Safety*, vol. 47, pp. 13–23, 2014.
- [8] K. Roscoe and A. Hanea, "Bayesian networks in levee system reliability," in *Proc. 12th Int. Conf. Applications of Statistics and Probability in Civil Engineering (ICASP12)*, Vancouver, BC, Canada, 2015.
- [9] G. B. Baecher and J. T. Christian, *Reliability and Statistics in Geotechnical Engineering*. Chichester, U.K.: Wiley, 2003.
- [10] G. Shafer, *A Mathematical Theory of Evidence*. Princeton, NJ, USA: Princeton University Press, 1976.
- [11] A. Karpatne, G. Atluri, J. H. Faghmous, M. Steinbach, A. Banerjee, A. Ganguly, S. Shekhar, N. Samatova, and V. Kumar, "Theory-guided data science: A new paradigm for scientific discovery from data," *IEEE Transactions on Knowledge and Data Engineering*, vol. 29, no. 10, pp. 2318–2331, 2017.
- [12] C. Rudin, "Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead," *Nature Machine Intelligence*, vol. 1, no. 5, pp. 206–215, 2019.
- [13] C. Elkan, "The foundations of cost-sensitive learning," in *Proc. 17th Int. Joint Conf. Artificial Intelligence (IJCAI)*, 2001, pp. 973–978.
- [14] C. K. Chow, "On optimum recognition error and reject tradeoff," *IEEE Transactions on Information Theory*, vol. 16, no. 1, pp. 41–46, 1970.
- [15] Y. Geifman and R. El-Yaniv, "Selective classification for deep neural networks," in *Advances in Neural Information Processing Systems*, vol. 30, 2017, pp. 4878–4887.
- [16] F. Wellmann and G. Caumon, "3-D structural geological models: Concepts, methods, and uncertainties," in *Advances in Geophysics*. Elsevier, 2018, vol. 59, pp. 1–121.
- [17] M. de la Varga, A. Schaaf, and F. Wellmann, "GemPy 1.0: open-source stochastic geological modeling and inversion," *Geoscientific Model Development*, vol. 12, no. 1, pp. 1–32, 2019.
- [18] L. A. Richards, "Capillary conduction of liquids through porous mediums," *Physics*, vol. 1, no. 5, pp. 318–333, 1931.
- [19] M. A. Celia, E. T. Bouloutas, and R. L. Zarba, "A general mass-conservative numerical solution for the unsaturated flow equation," *Water Resources Research*, vol. 26, no. 7, pp. 1483–1496, 1990.
- [20] M. T. van Genuchten, "A closed-form equation for predicting the hydraulic conductivity of unsaturated soils," *Soil Science Society of America Journal*, vol. 44, no. 5, pp. 892–898, 1980.
- [21] L. Moreau and P. Missier, "PROV-DM: The PROV data model," W3C Recommendation, Apr. 2013. [Online]. Available: <https://www.w3.org/TR/2013/REC-prov-dm-20130430/>
- [22] J. R. Quinlan, "Induction of decision trees," *Machine Learning*, vol. 1, no. 1, pp. 81–106, 1986.
- [23] L. Breiman, "Random forests," *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.
- [24] T. Hastie, R. Tibshirani, and J. Friedman, *The Elements of Statistical Learning*, 2nd ed. New York, NY, USA: Springer, 2009.
- [25] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Advances in Neural Information Processing Systems*, vol. 30, 2017, pp. 4765–4774.
- [26] I. Colomina and P. Molina, "Unmanned aerial systems for photogrammetry and remote sensing: A review," *ISPRS Journal of Photogrammetry and Remote Sensing*, vol. 92, pp. 79–97, 2014.
- [27] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, ser. LNCS, vol. 9351, 2015, pp. 234–241.
- [28] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 779–788.
- [29] R. Kumar, H. H. Stutz, and K. Johari, "Localized identification of seepage and ponding in earthen embankment using infrared thermography assimilated with different deep learning frameworks," *Scientific Reports*, vol. 15, 2025.
- [30] T. Saito and M. Rehmsmeier, "The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets," *PLoS ONE*, vol. 10, no. 3, p. e0118432, 2015.
- [31] K. Deb, A. Pratap, S. Agarwal, and T. Meyarivan, "A fast and elitist multiobjective genetic algorithm: NSGA-II," *IEEE Transactions on Evolutionary Computation*, vol. 6, no. 2, pp. 182–197, 2002.
- [32] S. Kaufman, S. Rosset, C. Perlich, and O. Stitelman, "Leakage in data mining: Formulation, detection, and avoidance," *ACM Transactions on Knowledge Discovery from Data*, vol. 6, no. 4, pp. 15:1–15:21, 2012.
- [33] D. R. Roberts, V. Bahn, S. Ciuti, M. S. Boyce, J. Elith, G. Guiller-Arroita, S. Hauenstein, J. J. Lahoz-Monfort, B. Schröder, W. Thuiller, D. I. Warton, B. A. Wintle, F. Hartig, and C. F. Dormann, "Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure," *Ecography*, vol. 40, no. 8, pp. 913–929, 2017.
- [34] Q. McNemar, "Note on the sampling error of the difference between correlated proportions or percentages," *Psychometrika*, vol. 12, no. 2, pp. 153–157, 1947.
- [35] T. G. Dietterich, "Approximate statistical tests for comparing supervised classification learning algorithms," *Neural Computation*, vol. 10, no. 7, pp. 1895–1923, 1998.
- [36] B. Efron and R. J. Tibshirani, *An Introduction to the Bootstrap*. New York, NY, USA: Chapman & Hall, 1993.
- [37] C. J. Clopper and E. S. Pearson, "The use of confidence or fiducial limits illustrated in the case of the binomial," *Biometrika*, vol. 26, no. 4, pp. 404–413, 1934.